

Original papers

RSHRNet: Improved HRNet-based semantic segmentation for UAV rice seedling images in mechanical transplanting quality assessment

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ARTICLE INFO

Keywords:

Unmanned aerial vehicle
Semantic segmentation
Rice transplanter
Smart agriculture

ABSTRACT

Rice seedling morphological identification is crucial for assessing the quality of mechanical transplanting, a process that has traditionally relied on subjective and inefficient manual inspections. To address this challenge, this paper introduces a non-contact quality assessment approach for rice mechanical transplanting based on unmanned aerial vehicle (UAV) imagery. Specifically, we propose a semantic segmentation model for rice seedlings based on an improved HRNet architecture, named RSHRNet. Leveraging UAV imagery as the data source, the methodology employs HRNet as the backbone network, facilitating the acquisition of high-resolution feature information through parallel interactions. Then, the object-contextual representations (OCR) module and coordinate attention mechanism are subsequently introduced to comprehensively aggregate contextual feature information and enhance the model's ability to extract spatial positional information. This approach helps to resolve issues of blurry edges and misclassification related to rice seedling target segmentation. Finally, to validate the effectiveness and advancements of the proposed method, comparative analyses are conducted against classical segmentation algorithms, followed by practical tests. Experimental results demonstrate the outstanding performance of the proposed RSHRNet in the segmentation of rice seedlings captured by UAVs, providing a solid foundation for the evaluation of mechanical transplanting quality.

1. Introduction

Rice is one of the world's three major staple crops, and its efficient production and stable quality directly influence global grain supply and food security. Due to the relatively high labor costs associated with manual transplanting, mechanical transplanting has been widely adopted as an alternative in many regions (Wang et al., 2021; Cui et al., 2023). However, a key challenge for mechanical transplanting is the intelligent detection of transplanting quality, which directly impacts the effectiveness and yield improvement of the process (He et al., 2020; Yang et al., 2020). To address this challenge, the introduction of sensing technology, image recognition techniques, and data analysis methods has provided potential solutions (Lohan et al., 2022; Li et al., 2024a). For instance, monitoring parameters such as transplanting depth, spacing, and posture can indirectly reflect the quality of transplanting machines (Ma et al., 2024a). Therefore, utilizing advanced technologies for quality assessment in the mechanical transplanting of rice holds significant implications.

Rice seedling morphological segmentation represents a critical technology for evaluating the quality of mechanical rice transplanting. The

acquisition of seedling posture information through image segmentation algorithms provides essential data for assessing transplanting quality (Ma et al., 2022b). In recent years, the widespread adoption of unmanned aerial vehicle (UAV) remote sensing technology in agricultural applications, including crop area identification, growth monitoring, yield estimation, and disaster warning, has generated substantial high-resolution aerial imagery, thereby establishing a robust data foundation for rice seedling image segmentation research (Rejeb et al., 2022; Li et al., 2024b). While traditional machine learning approaches, such as region-based, texture-based, threshold-based, and edge detection-based segmentation methods, have demonstrated certain achievements in image segmentation, they exhibit notable limitations when addressing the precise segmentation requirements of high-resolution UAV-based rice seedling images (Zhu et al., 2022). These limitations primarily manifest in inadequate multi-scale feature extraction capabilities, limited robustness, susceptibility to over-segmentation, and high computational complexity.

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Received 29 February 2024; Received in revised form 11 February 2025; Accepted 7 March 2025

Available online 22 March 2025

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Semantic segmentation technology has emerged as a superior solution for rice seedling morphological recognition due to its unique pixel-level classification capability. This advanced technique not only enables effective differentiation between rice seedlings and background elements but also facilitates precise identification of fine-grained features, including leaf morphology, dimensions, orientation, and stem positioning, thereby providing reliable morphological data support for transplanting quality evaluation. Compared to conventional object detection or image classification methods, semantic segmentation offers two distinct advantages: First, in complex field environments where rice seedlings are subject to interference from soil background, weed presence, and illumination variations, deep learning-based semantic segmentation models demonstrate enhanced robustness in morphological recognition through their powerful feature learning capabilities (Wu et al., 2023b). Second, the multi-scale feature extraction mechanisms employed in modern semantic segmentation architectures enable both accurate contour segmentation of individual seedlings and comprehensive analysis of spatial distribution patterns at a global scale, thus supporting quantitative evaluation of critical transplanting quality indicators such as planting density, uniformity, and morphological consistency.

Building upon these advantages, this study focuses on the application of UAV image semantic segmentation technology for rice seedling morphological information extraction and its subsequent utilization in transplanting quality assessment. By precisely segmenting leaf and stem regions, the proposed approach enables effective identification of distinct morphological characteristics of rice seedlings in soil (e.g., qualified seedlings, floating seedlings, and damaged seedlings), thereby facilitating indirect evaluation of mechanical transplanting performance. This research not only establishes a novel technical pathway for rice transplanting quality assessment but also contributes significantly to the advancement of precision agriculture practices.

1.1. Research review

There are primarily two categories of methods for assessing the quality of mechanical transplanting. The first category is the traditional method of random manual sampling testing (Zhu et al., 2022). This procedure involves personnel selecting several representative areas in the experimental field using a random sampling approach after transplanting. Subsequently, staff visually assess the morphology of seedlings, recording instances such as floating and injury. Despite the simplicity and lower technical requirements of this method, it faces several challenges: (1) High subjectivity, as the assessment of seedling morphology can be influenced by domain knowledge, leading to varying results among different personnel. (2) High randomness and considerable errors, primarily due to the difficulty in obtaining extensive data samples on a large scale, making it challenging for small-scale seedling tests to reflect the true performance of the machine. (3) Difficulty in tracing historical data due to the inability to systematically store morphological data during testing, making it challenging to reproduce results. Consequently, traditional manual observation and statistical evaluation methods are not only time-consuming and labor-intensive but also suffer from drawbacks such as low automation, difficulty in information sharing, and delayed result analysis.

The second category includes methods based on image processing, primarily involving object detection and semantic segmentation. For instance, in the realm of object detection, Yousef Asiri (Asiri, 2023) proposed a method using shark smell optimization for classifying rice seedlings. However, this method is limited to the binary classification of rice seedlings and cannot accurately capture other morphological information. Hu et al. (2020) introduced a technique using laser radar data to identify the trajectories of rice seedlings, facilitating the non-destructive driving trajectory needs of paddy field management machinery. Wang et al. (2021) presented a row vector grid-based method for recognizing rice seedling rows, constructing an

end-to-end CNN model that includes seedling feature extraction and row vector grid classification. This approach transforms seedling row recognition into a grid classification problem based on row vectors and global features. Wu et al. (2023a) proposed a rice field management navigation method based on seedling coordinate information. This method uses the YOLOv5s object detection algorithm to identify and locate rice seedling points in UAV images and establishes a coordinate transformation model to extract their geographical coordinates.

Furthermore, aiming for more accurate acquisition of multidimensional information about targets, scholars have explored image semantic segmentation. For example, Tian et al. (2024) proposed a method for crop area segmentation in remote sensing images using Deeplabv3+. This method segments regions including rapeseed, corn, fallow land, bare land, wheat, and rice. Addressing the challenges of high cost and uncertain annotation accuracy in dataset creation, Gao et al. (2023a) introduced a self-supervised semantic segmentation method for estimating Green Fraction (GF) in distinctly contrasting field images of rice, wheat, and weeds, thereby enhancing the accuracy of GF estimation in rice and wheat fields. Kumar et al. (2024) presented a semantic segmentation method for wheat rust disease (WRD) detection based on panoramic hybrid segmentation, used for real-time detection of WRD in images. Ma et al. (2020b) proposed a semantic segmentation method for rice seedling panicles, named EarSegNet, effectively segmenting rice seedling panicles from canopy images captured during the flowering stage. Furthermore, Lu et al. (2021) suggested an interpretable plant counting method, TasselNetV3, utilizing guided upsampling and background suppression. This method enables count redistribution through upsampling operators, using different fusion strategies of segmentation decoders to suppress the background. The experiments conducted included corn tassel counting, wheat panicle counting, and rice plant counting.

In summary, the aforementioned studies highlight the feasibility of using image processing techniques (Ma et al., 2024b, 2023) in crop morphology recognition. This approach can overcome the limitations of traditional manual assessment methods, which involve difficulties in automatic feature extraction, tedious and redundant processes, and challenges related to information sharing and delayed result analysis.

1.2. Main contributions and innovations

The previous studies have provided valuable references and a solid research foundation for quality assessment of mechanical rice transplanting using semantic segmentation. However, these technologies still face challenges with information loss during downsampling or upsampling. They often predict category labels solely from a pixel-level perspective without adequately considering the regional aspects of segmented objects, leading to suboptimal segmentation performance. Therefore, this paper, aiming to evaluate the quality of mechanical rice transplanting, proposes a rice seedling semantic segmentation model based on an improved HRNet model.

Given the complex land features in UAV images and the small, irregular spatial geometry of rice seedlings, this paper introduces an object-contextual representations (OCR) module (Yuan et al., 2020) based on the High-Resolution Network (HRNet) (Wang et al., 2020). This module is employed for coarse category-region segmentation of the output features from the backbone network, aiming to enhance the correlation between regions and pixels. Additionally, to extract clearer edge details, a coordinate attention mechanism (Hou et al., 2021) is introduced to strengthen pixel-to-pixel correlations. This mechanism enhances weight allocation for rice seedling features, optimizing the network's ability to extract detailed edge information from the target. As a result, the model can more accurately segment and extract small, irregularly shaped rice seedlings, thereby improving the assessment of mechanical transplanting quality. Therefore, the contributions and innovations of this study can be summarized as follows:

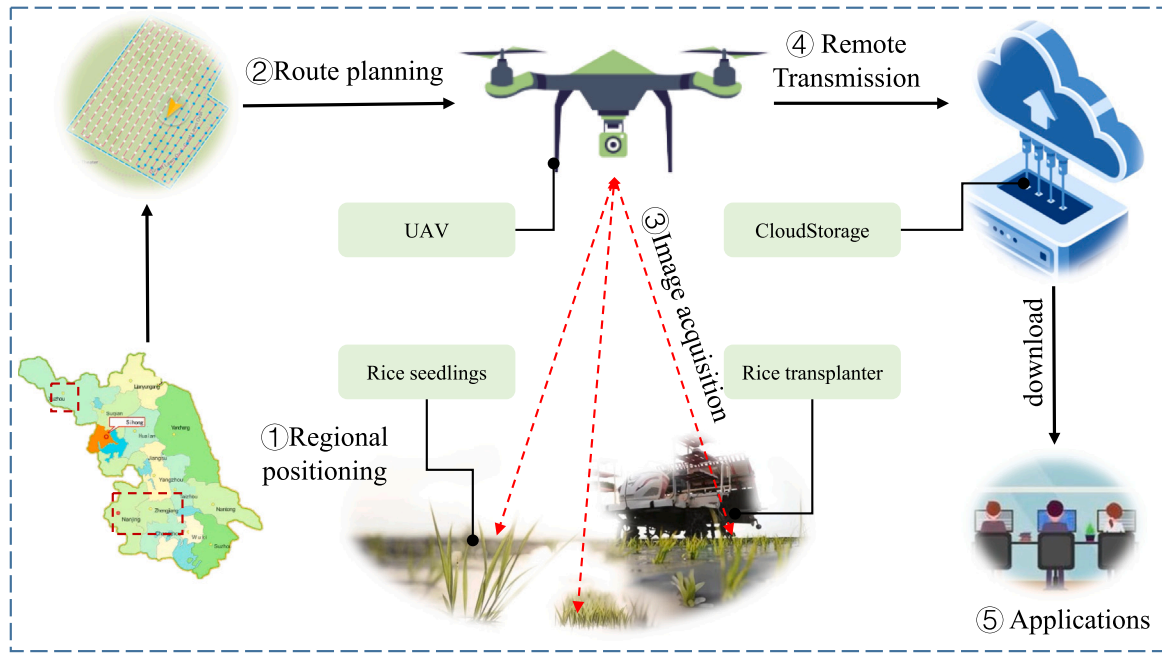


Fig. 1. Non-contact mechanical rice transplanting quality assessment scheme based on UAV images proposed in this paper.

- We proposed a non-contact quality assessment scheme for mechanical transplanting. To the best of our knowledge, this is the first attempt to move away from traditional reliance on manual inspection for assessing the quality of mechanical transplanting. We utilized UAVs to achieve non-contact morphological sensing of rice seedlings, thereby overcoming the challenges of subjectivity in manual judgment and labor-intensive tasks. More importantly, over the course of three years, our team established the first UAV-based rice seedling image dataset with sample diversity for mechanical transplanting quality evaluation. This dataset is expected to offer more research opportunities in rice breeding, crop protection, and food security.
- We present an enhanced HRNet model for semantic segmentation of aerial images. Leveraging HRNet as the backbone network, we employ a parallel interaction approach to capture high-resolution feature information. The joint module aggregates contextual information to strengthen the correlation between pixels and target object regions. Additionally, we introduce a coordinate attention mechanism to enable the network to fully leverage spatial position information effectively. This mechanism preserves edge details in segmented regions, thereby enhancing the segmentation accuracy for rice seedlings.
- We conducted extensive experimental validation. We validated the effectiveness and outstanding performance of our method on the UAV rice seedling dataset, demonstrating significant advantages over classical and the latest related methods, offering promising improvements for mechanical transplanting quality assessment and precision agriculture development.

1.3. Outline

The remaining of this paper is organized as follows: Section 2 describes the proposed UAV-based implementation of rice seedling image data acquisition work and explains the implementation details and key processes of the methods. Section 3 describes the experimental procedure and represents the comprehensive experimental results. Section 4 gives the conclusion of this study.

Table 1
Parameter and value settings of software and hardware for UAV rice seedling image acquisition.

Contents	Parameter or Value
UAV version	DJI phantom 4 RTK
Height	3 ± 0.5 m
Speed	1 ± 0.2 m/s
Flight times	3
Camera angle	90° ± 2°
Remote transmission library	DJI mobile SDK
Route planning software	DJI pilot

2. Materials and methods

This section introduces the implementation scheme for mechanical transplanting quality assessment based on UAV rice seedling images, as well as the improved semantic segmentation model (RSHRNet) for rice seedling images. The specific content includes image acquisition, method overview, OCR module, coordinate attention mechanism, and mechanical transplanting quality assessment.

2.1. Image acquisition

The rice seedling images used in this study were collected from May to June during the years 2019 to 2022, at experimental rice fields in Nanjing, Changzhou, and Liyang cities in Jiangsu Province, China. The rice seedlings were 15 days, 20 days, and 24 days old, representing the varieties Huaidao 5, RuanYu 2, and Nanjing 505, respectively. The transplanting machines used were Wode 2ZGF-6E, Fude 2ZG-8A, and Fengjiang 2ZG-4A. All images were collected during two time periods: 8:00 AM to 10:00 AM and 2:00 PM to 4:00 PM. Weather conditions varied mainly between overcast and sunny.

The UAV-based rice seedling mechanical transplanting quality assessment scheme includes several key steps, as shown in Fig. 1. Initially, in the selection of the UAV and flight parameters, a DJI Phantom 4 RTK, a compact professional-grade quadcopter, was chosen. It is equipped with a 12 MP CMOS camera and supports 4K video resolution (Liu et al., 2023). This device is capable of maintaining image detail even at considerable flight distances, ensuring the clear capture of crucial

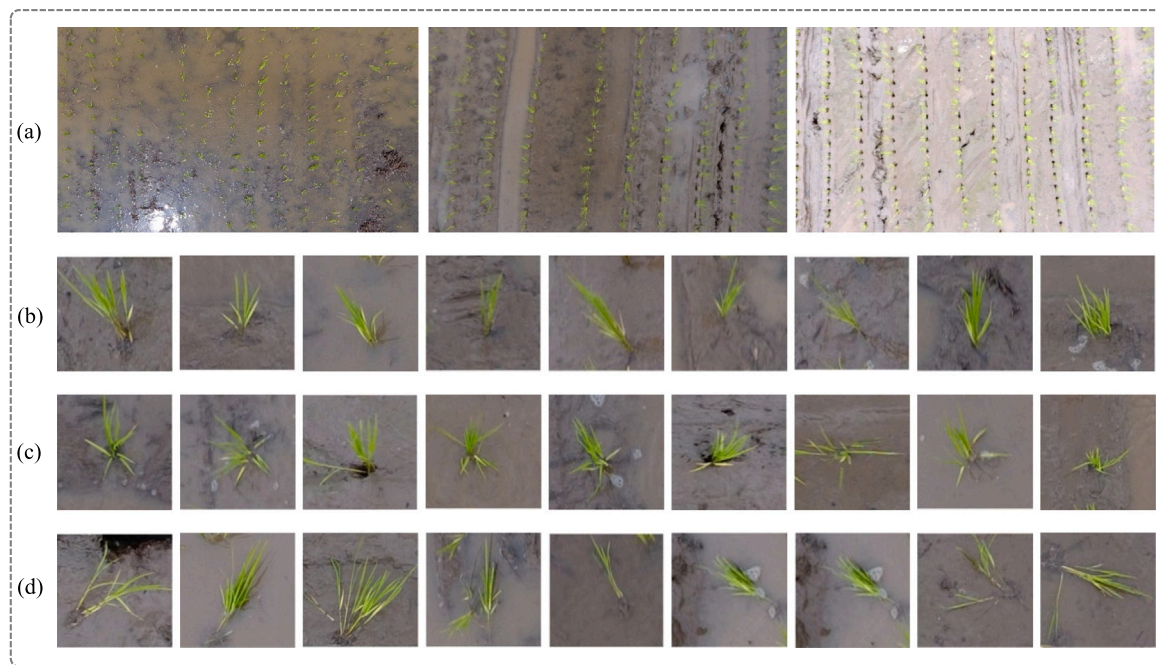


Fig. 2. Images and typical morphological samples of rice seedlings acquired by UAV: (a) Images obtained in different environments; (b) Qualified seedlings; (c) Damaged seedlings; (d) Floating seedlings.

features such as rice seedling leaves and stems, which is essential for subsequent semantic segmentation, especially under low-resolution conditions where target features must remain distinguishable.

In addition to the high-resolution camera, we further enhanced image clarity through the optimization of flight altitude, flight path planning, and camera parameters. Specifically, through repeated experiments, we identified the optimal flight height to ensure a wide field of view while maintaining an adequate pixel count for each rice seedling in the images. Furthermore, by integrating super-resolution image processing techniques, we enhanced the details of low-resolution images, thus improving the resolution of the rice seedling regions and ensuring that morphological features were sufficiently clear to meet the requirements for semantic segmentation.

Moreover, the UAV is equipped with an RTK positioning system, which provides centimeter-level positioning accuracy to ensure flight stability and maintain consistent image angles during the capture process. This minimizes the potential impact of motion blur or angular deviation on image quality and accuracy. Consequently, the UAV image acquisition process encompasses regional positioning, flight path planning, image capture, remote transmission, and post-processing. The relevant parameter settings are presented in Table 1. Ultimately, the rice seedling image samples acquired through this scheme are shown in Fig. 2.

2.2. Overview

HRNet is a deep learning network architecture originally designed for human pose estimation tasks in high-resolution images. Its primary goal is to efficiently capture multi-scale information while preserving high-resolution features (Wang et al., 2020). Unlike traditional deep learning networks (Ma et al., 2022a, 2020a), HRNet employs a unique high-resolution feature fusion strategy. Within the network, HRNet maintains multiple resolution feature maps and efficiently integrates these maps through a fusion mechanism. This enables the network to capture features at various resolutions more effectively, addressing the challenges associated with multi-scale information, and making it widely applicable in the field of image semantic segmentation.

However, the model is more suited for general images with minimal background interference and without significant differences (Wu et al.,

2021; Li et al., 2023). For UAV-captured rice seedling images with complex terrain information and similar features, where target pixels are small and boundaries are blurry, the model may be prone to interference. This interference can lead to partial loss of pixel details and inaccuracies in predictions. To address these issues, we propose an improved rice seedling HRNet, RSHRNet. The method overview is shown in Fig. 3, and the workflow is as follows:

The proposed RSHRNet is based on HRNet, integrating both the OCR module and coordinate attention mechanism. Unlike traditional networks with sequential downsampling and upsampling, RSHRNet adopts a parallel structure, leveraging HRNet as its backbone feature extraction network. This parallel connection of sub-networks, ranging from high to low resolution, deviates from the conventional approach and maintains high-resolution feature representations throughout the entire process. The intertwining connections of sub-networks at different resolutions, coupled with the fusion of representations generated by multi-scale sub-networks, effectively maintain feature information. This design enhances the network's ability to handle multi-scale information.

Furthermore, to address the challenges posed by complex backgrounds in aerial images and the potential confusion of target features, we integrate the main network with the OCR module. This integration aims to enhance the accurate extraction of feature information by aggregating the contextual features of rice seedlings. The OCR module utilizes target features extracted by the HRNet and performs weighted aggregation across all object regions in the image through supervised classification. This process achieves a coarse segmentation of the image, delineating K -class segmentation maps that distinctly outline the approximate positions of target objects. This provides robust positional information for subsequent processing.

Moreover, to address the issue of unclear boundaries of rice seedlings in UAV images, we introduce a coordinate attention mechanism within the OCR module. This implementation aims to improve the precise segmentation at the intersections of object regions with the indistinct edges of rice seedling areas. It enhances the relationship between object regions and corresponding pixels, leading to more accurate classification of rice seedling pixels. Ultimately, by relying on the precise segmentation results of rice seedlings, we can effectively

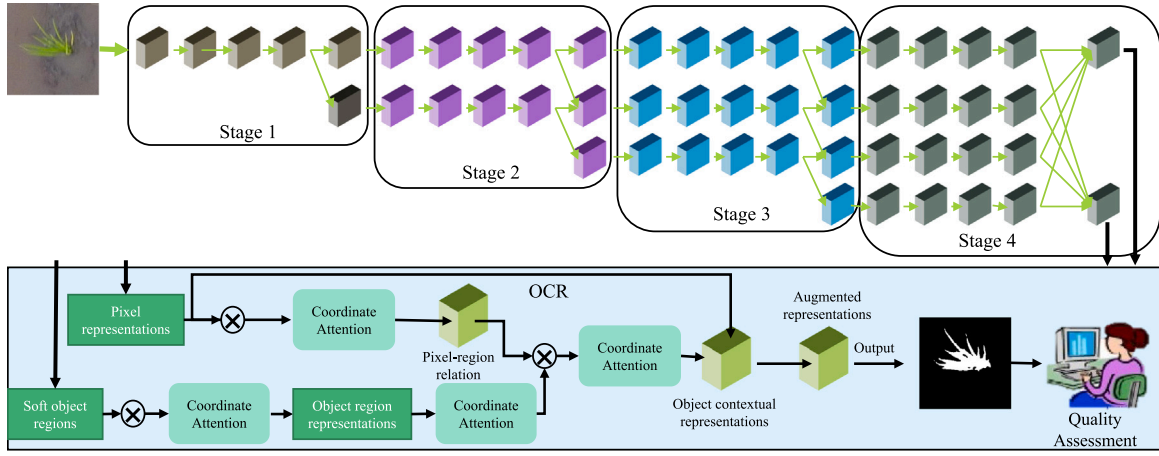


Fig. 3. The overall framework of the proposed RSHRNet method.

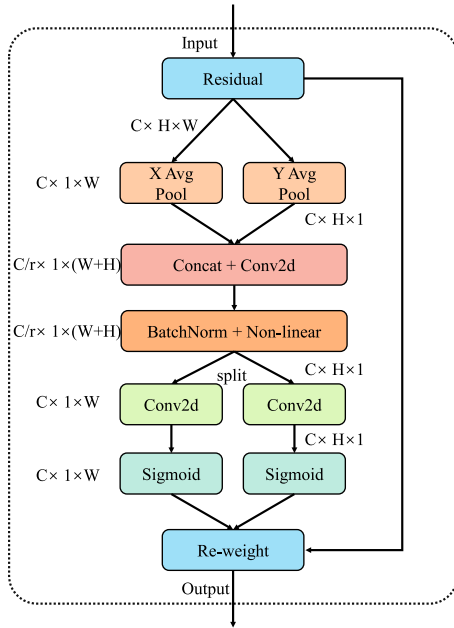


Fig. 4. Structure of the coordinate attention mechanism module. Here, “X Avg Pool” and “Y Avg Pool” refer to one-dimensional horizontal global pooling and one-dimensional vertical global pooling, respectively.

assess the quality of mechanical transplanting. This integrated processing workflow enables the model to better address the segmentation challenges of rice seedlings in complex environments. Consequently, it establishes a reliable foundation for further evaluations of mechanical transplanting quality.

2.3. OCR module

To enhance the modeling capability of relationships between objects and their context in images, the main concept of OCR lies in simultaneously attending to both the object and the surrounding contextual information to improve the understanding of objects in the image. Traditional visual models may solely focus on the object itself, whereas OCR, by incorporating contextual information, enables a more comprehensive understanding of the object’s context. This helps address ambiguities in complex scenes and enhances the accuracy of target identification. A brief description of the working steps and principles of OCR in semantic segmentation is outlined as follows:

Step 1. Image segmentation and category partitioning: For an image I with pixel coordinates (x, y) , the OCR module first performs weighted aggregation to obtain the segmentation map S_k for K classes (Yuan et al., 2020), representing the segmentation map of the k th class object.

$$S_k(x, y) = \sum_{i,j} \alpha_{(x,y),(i,j)}^k \cdot I(i, j). \quad (1)$$

Step 2. Pixel-region relationship representation: After identifying the positions of target objects, consider the relationship between pixel (x, y) and all classes of object regions, and perform feature-weighted aggregation to obtain the pixel-region relationship representation $R(x, y)$.

$$R(x, y) = \sum_k w_k \cdot S_k(x, y). \quad (2)$$

Step 3. Weighted aggregation of contextual information: For each pixel (x, y) , consider its feature representation $f_{(x,y)}$ and the pixel-region relationship $R(x, y)$, and perform weighted aggregation to obtain the enhanced feature representation $\hat{f}_{(x,y)}$ with contextual information.

$$\hat{f}_{(x,y)} = f_{(x,y)} + \beta \cdot R(x, y). \quad (3)$$

In these equations, $\alpha_{(x,y),(i,j)}^k$ represents attention weights, w_k represents weights for different categories, and β represents fusion weights. Through weighted aggregation, this process effectively extracts the spatial relationships between rice seedling pixels and other terrain pixels, enhancing the information representation of the foreground target, rice seedlings.

2.4. Coordinate attention mechanism

To address the issues of pixel loss and insufficient edge detail information during the downsampling and upsampling processes in the backbone network HRNet, and to enhance the acquisition of semantic information while improving the spatial features of target objects, we introduce the Coordinate Attention Mechanism module, as illustrated in Fig. 4. This module encodes channel relationships and distant dependencies precisely using accurate positional information. It is primarily composed of two key steps: coordinate information embedding and coordinate attention generation.

2.4.1. Coordinate information embedding

In the coordinate information embedding stage, the module incorporates precise positional information into the feature representation, introducing spatial location details for each pixel. This helps the model achieve a more accurate understanding of each pixel’s position in the image, thereby better capturing the spatial features of the target object (Hou et al., 2021). Specifically, we use pooling kernels of size

$(H, 1)$ to encode each channel along the horizontal coordinate direction and pooling kernels of size $(1, W)$ to encode each channel along the vertical coordinate direction of the input feature map. The spatial features from both directions are then aggregated to precisely locate the target object. Consequently, the output for the c th channel with height h and width w can be expressed as follows:

$$\begin{cases} z_c^h(h) = \frac{1}{W} \sum_{0 \leq i < W} x_c(h, i) \\ z_c^w(w) = \frac{1}{H} \sum_{0 \leq j < H} x_c(j, w) \end{cases} \quad (4)$$

2.4.2. Coordinate attention generation

In the coordinate attention generation stage, we use the embedded coordinate information to generate coordinate attention weights. These weights are used for the weighted fusion of feature maps, enabling the model to concentrate more specifically on spatially relevant features at particular locations (Hou et al., 2021). This process enhances focus on the target object and reinforces semantic information without introducing additional computational overhead. Specifically, both feature maps from the two spatial directions undergo a joint transformation using a 1×1 convolution, and can be represented as:

$$f = \delta(F_1([z^h, z^w])). \quad (5)$$

where, $f \in \mathbb{R}^{C/r \times (H+W)}$ represents the intermediate feature map after encoding operations in the horizontal and vertical directions. F_1 denotes the convolutional block after feature map transformation. δ represents the non-linear activation function. Subsequently, we split the feature map f along the spatial dimension into two independent tensors, $f^h \in \mathbb{R}^{C/r \times H}$ and $f^w \in \mathbb{R}^{C/r \times W}$.

Additionally, two additional 1×1 convolutional transformations, F_h and F_w , are applied to respectively transform f^h and f^w into tensors with the same number of channels as the input X , resulting in:

$$\begin{cases} g^h = \sigma(F_h(f^h)), \\ g^w = \sigma(F_w(f^w)). \end{cases} \quad (6)$$

where, σ represents the sigmoid function, and the outputs g^h and g^w are expanded and used as attention weights, respectively. Finally, the output of our coordinate attention block y can be expressed as:

$$y_c(i, j) = x_c(i, j) \times g_c^h(i) \times g_c^w(j). \quad (7)$$

Therefore, by introducing the coordinate attention mechanism, we further optimize HRNet's performance, particularly in handling pixel details and edge information, to achieve more comprehensive and accurate semantic information extraction. The combined effect of coordinate information embedding and coordinate attention generation provides the model with enhanced spatial awareness capabilities.

2.5. Loss function

In the task of rice seedling image segmentation, the loss function is a critical mechanism for evaluating the difference between model predictions and ground truth labels, directly affecting the model's convergence and segmentation accuracy. The loss function quantifies prediction errors, guiding the optimization of model parameters and ultimately improving segmentation performance (Chang et al., 2020). Commonly used loss functions include Cross-Entropy Loss, Dice Loss, and Intersection over Union (IoU) Loss. The Cross-Entropy Loss (L_{CE}) function is widely used in both multiclass and binary classification problems. It estimates error by measuring the difference between the predicted distribution and the true distribution. Its mathematical expression is:

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]. \quad (8)$$

where N represents the total number of samples, y_i denotes the true label of the i th pixel, and $\hat{y}_i$ represents the probability of that pixel

being predicted as the positive class. By averaging the errors across all pixels, the Cross-Entropy Loss provides a global error metric for model training.

However, the traditional Cross-Entropy Loss function has limitations when handling imbalanced data and easily classified samples, potentially leading the model to overlook hard-to-classify samples, thereby affecting overall performance. To address this issue, this paper employs the Online Hard Example Mining (OHEM) Cross-Entropy Loss function. OHEM dynamically selects hard examples with higher loss values during training to compute the loss, enhancing the model's learning ability for these samples. The form of the loss function is:

$$L_{OHEM} = \frac{1}{v} \sum_{i \in S_v} L_{CE}(i). \quad (9)$$

where S_v represents the v hard examples selected from the current batch, and $L_{CE}(i)$ denotes the Cross-Entropy Loss value of the i th sample.

2.6. Quality assessment for transplanter

Based on the predictive results from RSHRNet, and in accordance with the standards outlined in "GB/T 6243-2017 Test Methods for Rice Transplanter" and "DG/T 008-2019 Rice Transplanter" (Zhu et al., 2022), we categorize the rice seedlings within the mechanical transplanting holes into floating seedlings, damaged seedlings, and qualified seedlings based on their morphological characteristics, defined as follows:

Definition 1 (Floating Seedlings). Seedlings whose roots are not planted in the soil within the transplanting hole but instead float on the surface of the paddy water.

Definition 2 (Damaged Seedlings). Seedlings with observable injuries, breakages, or cuts on the stem within the transplanting hole.

Definition 3 (Qualified Seedlings). Seedlings within the transplanting hole that do not exhibit floating, injuries, breakages, or cuts are considered qualified.

Therefore, in this study, the rates of floating, damaged, and qualified seedlings are calculated as follows:

$$\begin{cases} R_F = \frac{X_F}{X} \times 100\% \\ R_D = \frac{X_D}{X} \times 100\% \\ R_Q = (1 - \frac{X_F + X_D}{X}) \times 100\% \end{cases} \quad (10)$$

where R_F represents the floating seedling rate, X_F represents the total number of holes with floating seedlings, X represents the total number of measured seedling holes, R_D represents the damaged seedling rate, X_D represents the total number of holes with damaged seedlings, and R_Q represents the qualified seedling rate.

3. Experiments and results analysis

To evaluate the proposed method, this section provides a detailed report of the multiple validation experiments conducted on our self-constructed UAV-based rice seedling dataset. These experiments include the standards followed, implementation details, the design of comparative experiments, and the analysis of robustness test results.

3.1. Experimental settings

3.1.1. Dataset description

Based on the rice seedling data acquired by UAV imagery, and considering the intricacies of paddy field images and the demands of deep learning on input data, we conducted data preprocessing. First, the images were systematically cropped and divided into 256×256 pixel blocks. Subsequently, three experts in the field of mechanical transplanting evaluated the dataset using Labelme software for precise annotation, generating a label dataset that corresponds one-to-one with the original dataset in terms of order, quantity, and size. Notably, to enhance the model's robustness, several image enhancement techniques were employed during data preprocessing and augmentation. These included denoising, contrast enhancement, and edge enhancement, which significantly improved image clarity and detail discernibility. Such methods were particularly effective in retaining the morphological features of rice seedlings in complex backgrounds and under low-contrast conditions. Ultimately, a dataset containing 5200 samples was created and divided into training, validation, and test sets in an 8:1:1 ratio for model training, tuning, and performance evaluation.

3.1.2. Software and hardware configuration

To validate the proposed RSHRNet method in this study, experiments were conducted on a desktop computer equipped with an Intel Core i7-4790k CPU, 64 GB RAM, and an NVIDIA Titan X GPU. Parallel computation was performed using CUDA 11.7, with Python chosen as the programming language. The design, training, and tuning of the neural network model were conducted using the PyTorch deep learning framework. To effectively optimize model parameters during the training phase, the SGD algorithm was chosen as the optimization method, with an initial learning rate set to 0.01, weight decay at 0.0005, momentum set to 0.9, and a total of 300 iterations.

3.1.3. Evaluation metrics

To assess model performance, we utilized Mean Intersection over Union (MIoU) and Pixel Accuracy (PA) as evaluation metrics (Chen et al., 2018), conducting accuracy evaluations on the results of different models. MIoU is employed to measure the ratio of the intersection to the union of the sets between the ground truth and predicted values. Its calculation formula is as follows:

$$MIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{N_{ii}}{\sum_{i=0}^k N_{ij} + \sum_{j=0}^k N_{ji} - N_{ii}} \times 100\%. \quad (11)$$

PA is the ratio of the number of correctly labeled pixels to the total number of pixels and is calculated as follows:

$$PA = \frac{\sum_{i=0}^k N_{ii}}{\sum_{i=0}^k \sum_{j=0}^k N_{ij}} \times 100\%. \quad (12)$$

where $k+1$ is the number of classes, including the background class; N_{ii} is the number of pixels where pixel class i is correctly predicted as class i ; N_{ij} is the number of pixels where pixel class i is predicted as class j ; and N_{ji} represents the number of positive instances of incorrect prediction as well as the number of negative instances of incorrect prediction.

3.2. Results and analysis

3.2.1. Ablation studies

To evaluate the effectiveness of the OCR module and the Coordinate Attention Mechanism in improving model performance, three ablation experiments were conducted, followed by a comprehensive evaluation under consistent testing conditions. The experimental results, as shown in Table 2, indicate the presence of each module with a checkmark “✓” and its absence with a cross “×”. The analysis of these results clearly demonstrates the impact of different modules on segmentation performance. Initially, the HRNet baseline model was used as a

Table 2

Quantitative experimental results of ablation studies.

HRNet	OCR	Coordinate attention	MIoU (%)	PA (%)
✓	×	×	57.45%	90.27%
✓	✓	×	60.27%	91.49%
✓	✓	✓	62.68%	92.17%



Fig. 5. Visualization results of ablation experiments.

reference, achieving a MIoU of 57.45% and a PA of 90.27%. These results highlight the strong feature extraction capabilities of HRNet. Although the model exhibited satisfactory pixel-level classification for rice seedling segmentation, it showed somewhat blurred boundaries, with limited detail preservation.

Subsequently, the OCR module was incorporated, primarily to enhance the association between class regions and target pixels. This is particularly beneficial for fine-grained segmentation tasks involving complex backgrounds, where OCR can leverage contextual information to assist the model in better delineating boundaries between the target and background. Experimental results show that the HRNet+OCR combination improved MIoU and PA by 2.8% and 1.22%, respectively, compared to the baseline HRNet model, confirming the OCR module's effectiveness in improving pixel-wise associations.

Further improvement was achieved by introducing the Coordinate Attention Mechanism, which adapts the attention of the model to the spatial relationships between target pixels and their surrounding pixels. This mechanism enhances the model's sensitivity to complex spatial structures by performing fine-grained spatial weighting adjustments. It helps the model better capture and distinguish subtle differences between similar objects, especially at object contours and boundary regions. The results indicate that the HRNet+OCR+Coordinate Attention combination outperforms the HRNet baseline by 5.23% and 1.9% in terms of MIoU and PA, respectively. Additionally, compared to the HRNet+OCR combination, MIoU and PA were improved by 2.43% and

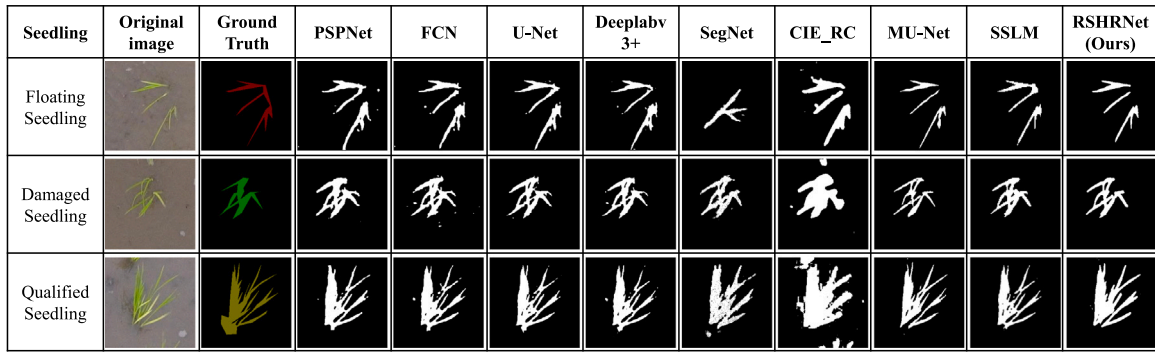


Fig. 6. Visualization results of different methods for segmentation of rice seedlings on the test set.

Table 3

Experimental comparison results of different methods on the test set, the results of this paper are shown in bold.

Methods	MIoU (%)	PA (%)	Parameters (MB)
PSPNet	55.26%	90.35%	48.97 MB
FCN	50.73%	91.74%	49.49 MB
U-Net	53.68%	89.73%	58.69 MB
Deeplabv3+	57.88%	90.58%	43.69 MB
SegNet	57.49%	89.63%	115.13 MB
CIE_RC	26.95%	74.56%	–
MU-Net	62.25%	90.77%	47.88 MB
SSLM	62.21%	91.49%	30.41 MB
RSHRNet	62.68%	92.17%	65.67 MB

0.68%, respectively. These findings suggest that the Coordinate Attention Mechanism not only boosts the effectiveness of the OCR module but also contributes to substantial overall performance enhancement.

For a more intuitive understanding of the ablation experiment outcomes, segmentation results from different modules were visualized. Fig. 5 illustrates the segmentation of various rice seedling shapes using different methods. The results indicate that the HRNet baseline model produced blurry boundaries with poor detail retention. In contrast, the inclusion of the OCR module resulted in clearer segmentation boundaries with improved detail preservation. The introduction of the Coordinate Attention Mechanism further refined the segmentation, ensuring more accurate boundary delineation and enhanced feature retention, particularly at the seedling edges, thus reducing the loss of critical information and improving segmentation accuracy. This enhancement enables the model to not only more accurately identify targets but also effectively suppress background noise, thereby enhancing segmentation performance.

In summary, the results from the ablation experiments and visualizations demonstrate that the HRNet+OCR+Coordinate Attention Mechanism combination offers a more comprehensive and significant performance improvement compared to individual module combinations. By integrating the OCR module and the Coordinate Attention Mechanism, our model substantially improves segmentation accuracy, particularly in complex environments, showcasing superior fine-grained segmentation capabilities for rice seedling images.

3.2.2. Comparative analysis with other methods

To thoroughly investigate the performance differences between the proposed RSHRNet and other methods in rice seedling segmentation, we conducted comparative experiments with several classical methods and recent advancements in the field, including PSPNet (Zhao et al., 2017), FCN (Long et al., 2015), U-Net (Ronneberger et al., 2015), Deeplabv3+ (Chen et al., 2018), SegNet (Badrinarayanan et al., 2017), CIE Lab* color space and region-based crop (CIE RC) (Kim et al., 2023), MU-Net (Zhang and Zhang, 2023), and Self-Supervised Learning Model (SSLM) (Gao et al., 2023b). Comprehensive experimental results are presented in Table 3.

By comparing the performance metrics of various models, we found that the proposed RSHRNet achieved the best performance, with an MIoU of 62.68% and a PA of 92.17%. Compared to SSLM (MIoU of 62.21% and PA of 91.49%), our MIoU improved by approximately 0.5% and PA by about 0.7%. The superior performance of RSHRNet can be attributed to its improvements in addressing the insufficient feature information problem in traditional segmentation models. We introduced the OCR module to enhance the correlation between region and target pixels, and incorporated the Coordinate Attention mechanism to strengthen pixel-to-pixel correlations, thereby capturing more precise detail information. In contrast, SSLM is based on a self-supervised learning approach that uses CycleGAN to effectively reduce domain gaps. However, significant differences remain between synthetic and real images, particularly in leaf texture, background, and lighting distribution, which may limit its effectiveness in rice seedling segmentation in this study. Additionally, we found that the CIE_RC-based method performed poorly. The primary reason is that CIE_RC employs the a^* channel in the CIE Lab* color space combined with the mean shift (MS) algorithm, which may be effective for larger targets and more distinct color regions, but performs poorly in the context of rice seedling segmentation in this study. However, despite the excellent performance of our RSHRNet, its parameter count has also increased. In practical applications, balancing the trade-off between high performance and parameter count is reasonable.

Fig. 6 presents the visualization results of rice predictions using different algorithms. We observed that these methods exhibited varying shape differences, with RSHRNet, MU-Net, and SSLM achieving better edge detail extraction in rice seedlings, while the CIE_RC-based method could only capture blurred contour information. In comparison, our proposed RSHRNet model demonstrated the best performance in edge detail extraction and feature recognition. This is because we fully considered the correlation between pixels and category regions, ensuring not only the integrity of the rice seedling feature information but also the precision of edge details. Therefore, our experimental results provide strong support for further research into the quality assessment of mechanical rice transplanting.

3.2.3. Robustness analysis

To evaluate the robustness of our model in complex scenarios, we conducted additional testing experiments under varying lighting and motion blur conditions. We introduced lighting and motion blur disturbances of varying intensities into the rice seedling test set to assess the model's resistance to interference. Lighting intensity is represented by ℓ ; the larger the ℓ , the stronger the lighting and the higher the image exposure. In the experiments, we set ℓ to four levels: -60 , -30 , 30 , and 60 . When ℓ is set to -60 (low light) and 60 (strong light), target extraction is usually challenging. Additionally, motion blur disturbance is represented by ∂ , where a larger ∂ indicates stronger disturbance and a more blurred target. In the experiments, we set ∂ to four levels (2, 4, 6, 8) to better analyze the model's robustness.

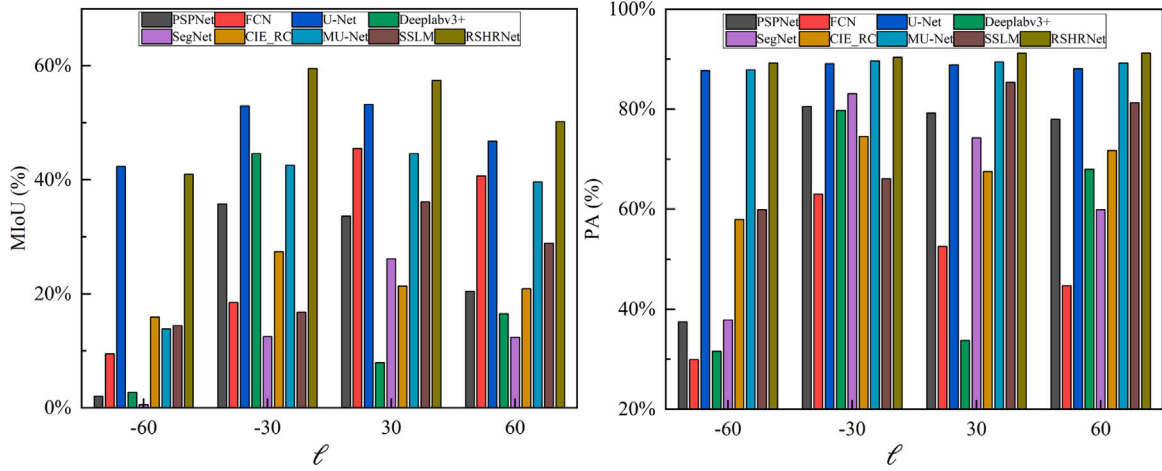


Fig. 7. Robustness test results of individual models under different light intensity conditions.

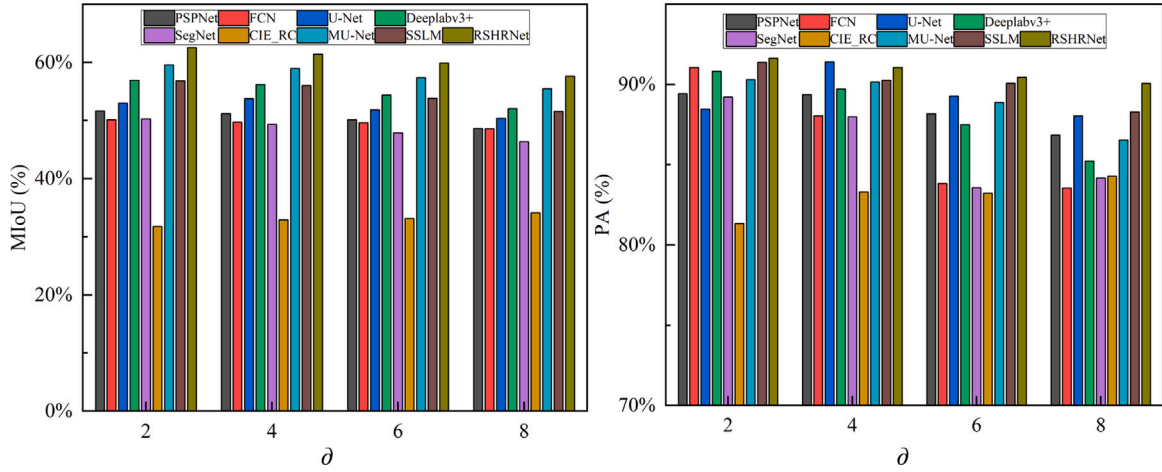


Fig. 8. Robustness test results of individual models under varying motion blur disturbance conditions.

Fig. 7 presents the results of the model robustness tests under different lighting levels. We observed that as the lighting intensity increased, each model exhibited varying degrees of change in the two evaluation metrics (MIoU and PA). Notably, all models were most affected under low light (-60) and strong light (60) conditions. However, the methods based on RSHRNet and U-Net demonstrated good robustness. Overall, our method maintained stable MIoU and PA outputs under the four different lighting conditions, likely due to HRNet's strong feature extraction capabilities and the introduction of the coordinate attention mechanism, which enhances pixel-to-pixel correlation and helps mitigate the impact of varying lighting intensities on the model.

Furthermore, we introduced motion blur samples of varying intensities into the rice seedling test set to study the model's resilience in complex scenarios, as shown in Fig. 8. We found that slight blur disturbances ($\delta = 2$, $\delta = 4$) had a relatively minor impact on most models trained through learning. Additionally, we observed that when the motion blur intensity was higher, our RSHRNet and MU-Net models remained relatively stable, with RSHRNet outputting higher MIoU and PA in most cases. This suggests that by using HRNet as the base architecture and introducing OCR and coordinate attention mechanisms, the network effectively utilized spatial positional information to preserve edge details in segmented regions, which also plays a positive role in resisting motion blur interference in complex environments.



Fig. 9. The experimental scene of mechanical transplanting seedling quality assessment using traditional scheme.

3.3. Applications

To verify the practical application of the proposed method in mechanical transplanting quality assessment, field experiments were conducted using two different experimental schemes: manual sampling (traditional scheme) and the UAV-based approach proposed in this paper (UAV scheme). The manual sampling followed the procedures

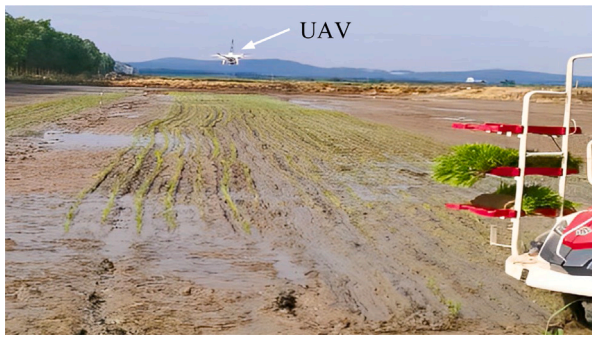


Fig. 10. The experimental scene of mechanical transplanting seedling quality assessment using UAV scheme.

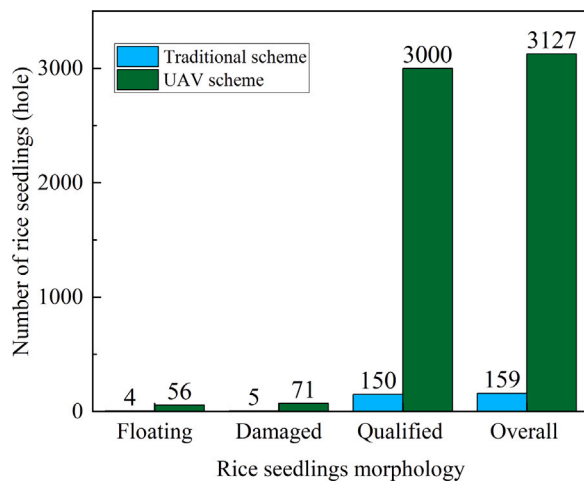


Fig. 11. Results of rice seedlings sample collection in the same time period by traditional scheme and UAV based scheme in this study.

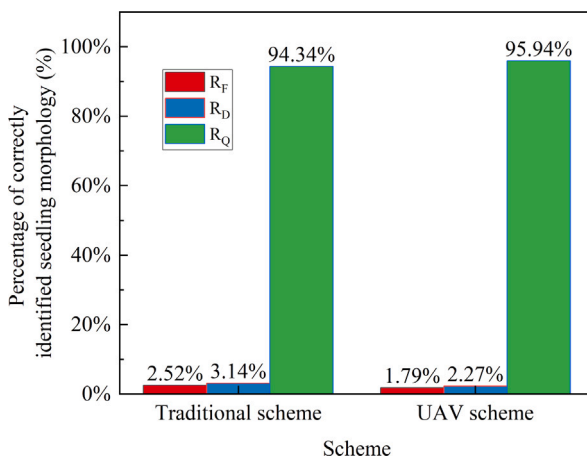


Fig. 12. Results of experimental comparison between UAV scheme and traditional scheme for quality assessment of mechanical rice transplanting.

outlined in the “GB/T 6243-2017 Rice Seedling Machine Test Methods” (Zhu et al., 2022). The experimental environment had a temperature of 27 °C, relative humidity of 70%, a rice field area of 97 × 60 square meters, clay soil type, water depth ranging from 3 to 5 cm, mud foot depth between 17 and 19 cm, and the seedlings used were manually cultivated Huaidao 5, with a seedling age of 20 days.

Firstly, in terms of rice seedling sample collection efficiency, the proposed method demonstrated significant advantages over the traditional scheme. In the traditional manual detection method, three staff members were involved in collecting seedlings, with one person recording the experimental data, as shown in Fig. 9. Due to the difficulty of covering large areas of rice fields with traditional scheme, sampling was conducted using a diagonal sampling method across five measurement zones, each spaced more than one working width apart. Within each zone, the staff measured indicators such as seedling count per hole, damaged seedlings, and floating seedlings. This process lasted between 20 and 25 min, and a total of 159 rice seedling holes were sampled. In contrast, the UAV scheme proposed in this paper, shown in Fig. 10, completed the collection of over 3000 rice seedling holes in just approximately 3 min by autonomously planning the flight path and using the UAV camera (Fig. 11). This significantly reduced the manual labor intensity and markedly increased sampling efficiency.

To further validate the effectiveness of the proposed method in mechanical transplanting quality assessment, Fig. 12 presents the experimental results of rice seedling morphology detection for both schemes. The experimental data shows that the proportion of qualified rice seedlings was the highest, while the proportions of floating seedlings and damaged seedlings were relatively low. In both detection schemes, the traditional scheme showed a slight advantage in detecting floating and damaged seedlings, with a floating rate of 2.52%, a damaged seedling rate of 3.14%, and a qualified seedling rate of 94.34%. In contrast, the UAV scheme proposed in this paper achieved a floating rate of 1.79%, a damaged seedling rate of 2.27%, and a qualified seedling rate of 95.94%. It is evident that the UAV method reduced the floating rate and damaged seedling rate by 0.73% and 0.87%, respectively, with relative errors controlled within 1%, which sufficiently meets the requirements for transplanting machine quality evaluation.

It is noteworthy that the traditional scheme faces issues of low efficiency and high labor costs, and due to the limited sample size, it may be subject to some subjective bias. On the other hand, the non-contact UAV scheme proposed in this paper significantly improves data collection efficiency, achieving approximately 6 to 8 times the efficiency of the traditional method, and offers advantages such as historical data retrieval. Therefore, the proposed method not only demonstrates high efficiency in data collection and processing but also provides more comprehensive and objective information for the experimental results, enhancing the reliability of rice transplanting machine quality evaluation. Additionally, this method greatly reduces manual labor intensity, offering an innovative solution for the mechanical quality assessment of rice transplanting.

3.4. Discussion

To enhance the automation level of rice mechanical transplanting quality assessment, this paper proposes a non-contact assessment scheme based on UAV images and designs a rice seedling segmentation method (RSHRNet). Although the strategy has achieved satisfactory results on the rice seedling dataset, there is still room for improvement. For instance, in real-world environments, rice seedlings with shorter cultivation cycles or those in complex backgrounds may affect model performance. While the introduction of the OCR module and coordinate attention mechanism has mitigated these issues to some extent, the model’s performance still holds potential for further enhancement. Additionally, the robustness and security of the model require further research. Under certain typical disturbances, such as variations in lighting intensity and motion blur, the proposed method has demonstrated good robustness. However, given the importance of rice information in food safety, the model’s ability to handle more potential disturbances (e.g., internal model disturbances and external attacks) warrants in-depth exploration. Specifically, the robustness of rice seedling image segmentation in the face of both internal and external disturbances

needs further investigation. Furthermore, this paper focuses on segmenting three typical rice seedling morphologies—qualified seedlings, damaged seedlings, and floating seedlings—to assist mechanical transplanting machine quality evaluation. However, the parameters involved in mechanical transplanting machine quality assessment may be more complex. Future research could consider expanding the evaluation metrics and delve deeper into other factors that influence quality assessment.

Therefore, future work could focus on several potential solutions. For example, further optimization of the internal structure of the HRNet network, combined with the OCR module and coordinate attention mechanism, could enhance the positioning accuracy and morphological expression of rice seedlings, thereby improving the model's feature extraction and segmentation performance. Additionally, addressing the model's robustness issue, research could approach the problem from both the UAV image data and segmentation model perspectives by simulating more types of disturbance samples and incorporating adversarial training to improve the model's resistance to various interferences. This would contribute to the development of a safer and more stable UAV-based rice transplanting quality assessment system, thereby advancing food information security and agricultural intelligence.

4. Conclusion

In this paper, we propose a non-contact quality assessment approach for rice mechanical transplanting using UAV imagery and semantic segmentation technology. Initially, diverse datasets of rice seedlings from mechanical transplanting under various scenarios were established through UAV aerial photography. Subsequently, an improved model based on HRNet was developed (RSHRNet). By incorporating joint OCR and coordinate attention mechanisms, the model effectively extracts contextual feature information from images and spatial relationships of the targets. This addresses challenges such as indistinct boundaries and misjudgments prevalent in rice seedling segmentation tasks. To validate the proposed method, extensive experiments and field tests were conducted. The experimental results demonstrate significant improvement over traditional manual detection methods. Furthermore, the introduced RSHRNet exhibits outstanding performance in segmenting rice seedling images captured by UAVs. In future research, we will further expand the evaluation metrics for mechanical transplanting machines and enhance the model's lightweight design and robustness to better handle more complex application scenarios.

CRedit authorship contribution statement

Xinghong Yang: Writing – original draft, Software, Methodology. **Heqing Li:** Methodology, Funding acquisition. **Wei Zhu:** Formal analysis, Data curation. **Yi Zuo:** Writing – review & editing, Funding acquisition, Data curation.

Declaration of generative AI and AI-assisted technologies in the writing process

None

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Jiangsu Province Agricultural Machinery New Equipment and New Technology R&D and Promotion Project (No. NJ2019-25). We would like to extend our sincere gratitude to Mr. Lixin Ma and his colleagues for their invaluable support in the collection of the rice dataset. Special thanks are due to Mr. Deying Liu and his team at the College of Engineering, Nanjing Agricultural University, for their critical assistance in the experimental validation. Additionally, all authors are deeply appreciative of the editors and reviewers for their constructive comments, which have significantly enhanced the quality of the manuscript.

Data availability

Data will be made available on request.

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